Data Science
##
Manifesto
    wrangle cyclopean tentacled chaos
    tame C'Thu'Lu to domestic service

Data easily becomes overwhelming.
Alien character compositions establish a barrier insurmountable
    -- to all but the faithful intelligent.

Dirty data is often a clean structure malformed for the current readership.

The first mission is to transform data into a human readable form.

Plan of action:
    What is the end purpose, expected-output, dimension?
    Who created the data: human or automation?
    Where was the data encoded; can encoding provide structural information?
    When was the time of creation; can time enable segregation?
    Why was the data created and how was it created? Does such reason match
your purpose?

WORKFLOW:
    1. Purpose
    2. Data source & data elements
    3. Sanatize data and discover data structure
    4. Explore & Analyze data
    5. Algorithm to structure data
    6. Verify results
    7. Reports
##
Remember

When the task is unsurmountable. Breathe. The first step is to be passive.
Watch the data. Print the data, element by element at a readable pace.
Let it become familiarized in the background.
Slices of data from random, start, mid, end - let a relationship develop.

First, one must be well-acquainted, and that takes time.
Patiently view the data while the system is designed.

Afterwards, the structure will be revealed.
Global dimensions which are able to contain any particular object.
Hierarchy to be flattened by classification.
Quarantine unstructured anonmaly to be analyzed last.
Work with objects that can be easily contained, and grow from that stable
center.
Do not try to contain all under universal dominion.

Identify patterns and relationships within the data.
Look for trends, correlations, and anomalies.
Apply operations to extract insights by trial & error.

Finally, apply appropriate techniques to extract insights and achieve the end
purpose.

Never forget that automation is the endgame.
Hands-on data grooming is essential until the system is mastered.
##
Purpose

Establish the smallest working set, of actions requiring interaction, by means of automata
    1) Read
    2) Reduce
    3) Transform
    4) Standardize
    5) Verify
    6) Automata
##
Core Ideas

FILTERS:
    Automata discrimination structures - to transform, join, exclude
        With the object to Sanitize & Standardize.
    Gatekeepers of what may be considered as input to the system.
    Logs & isolates unacceptable inputs.

POOLS:
    Collections of standardized objects of a final system.
    The trusted-working source for the system.

MAPS:
    Explicit relationships designed for operations, amongst elements & structures.
    Maps are built as the byways of pools to structure.

STRUCTURES:
    Preliminary transitional forms requisite for the system.
    Final form used by the system to compose results.
##
Element Reduction

Goal is to reduce the element into its irreducible unique form by operations which:
    { prune, compress, encode }

First analyze the whole, to obtain a minimum-structure which all fulfill.

Next consider the nature of the individual within the whole dimension
    Seek ways to operate towards reduction.

Never colapse elements into collision of identity
    Each element must remain unique.

If several sub-components can be collapsed into a single classification
    Then supplant those components with a single encoding.
    Encoding should be an essential distinction.

Nurture the virtue of Simplification, Clarity, & Efficency.
Redundancy is a burden; Complexity obfuscates unique natures.

##
Data Migration

Data often exists in an origin encoding
    First it must be extracted & preprocessed
    Afterwards filters can sanitize & standardize

Proprietary exports are hostile territories.
The major effort is to transform the data into a workable pool
    Without loss of information.
##
MAJOR BRANCHES OF SPECIALIZATION

    Encode: mutating external data to internal data AND internal data to external data
        Master common encodings to maximize structural information obtained form original.

    REGEX: internal data filters & substitution; builds intermediary data structures
        Magic of trasmuting data by automated filters
            At cost of extreme difficulty towards mastery.
        Endgame to master human & machine data transmutation.

    Transforms: internal data & final data structures
        Process sanitized data into operational constructs.
        Endgame is the architecture of information systems.

    Display: export the data to external data as list, table, graph, summary, report
        Humans eat with their eyes.
        Sketch captures structure in the most sophisticate manner
            Only oil painting is a craft of fine art.
##
CODING THEORY

LOGIC:
set
sequence
group

OPERATION:
if then
for
while
switch
matrix
combinatronics
remaindering (chisel towards)

STRUCTURE:
cell
array
hash
array of arrays

hash of arrays of arrays
struct
multi-set ledger
nested arrays, arrays with related sub-arrays

Dimension Operations:
	Distributive Expansion by Combinatronics: project vectors into higher dimensional structures.
	Reductive Collapse: shrinking dimensions.
		A flat vector is non-nested; relation only exists between the basis position & vector position (non-horizantal dependent relations).
		A flat vector can support any form of nested heirarchal data.
		Nested structures are designed for presentation.
		Flat, non-nested, vectors are ideal for storage & transmission.
		Row dimensional collapsing > column dimensional growth.
	Flattening: nested structures flattened into one vector per entry.
		Flattening often creates a sparse matrix.
		Decoupling of unrelated horizantal-relationships - relations must remain only basis unto entry.

#########
DATA SCIENCE

PURPOSE:
comparison
reduction
filtering
sanitizing
encodings
sorting
aggregation

############
PARSE
    Issues that arise from the data encodings are solved in the shell
    Issues that arise from internal data elements & structure resolved by programming
    The craft into the final is the domain of Data Analysis

These three layers of Data Science are kept seperate.
Encodings are the most difficult to wrangle.
Internal data works within a tame enviornment.
Data Analysis should be liberated from all issues not arising from the nature of the study.

ENCODING
Unix Terminal for Heterogenous Data Issues
awk, sed, ...
Domain is the first layer: what operating system created the data.
Platform is the second layer: what exact encoding created the data.
Destination format is the goal of the new enviornment to host the data.

What are the differences between original-encoding and destination-format?
The unix terminal unwraps the data entirely
    Independent from where & how it was created.
The original encoding forms the first data structure.
Use the filters to remove the old structure AND set the data into its first intermediary structure.

INTERNAL STRUCTURE
Code the system where the data will reside.
What forms the elements.
What is the atomic irreducible sub-element.
What is the basic unit.
What is the final struture required for the output.
What are the structures required to get the data into the final form.

INTERNAL ELEMENTS
REGEX for Internal Data Issues
pattern theory
    unique tag identifiers (pattern match target)
search
match
replace
rearrange
mutation
concatenate
anchors
net capture
formulae
chaining

OUTPUT:
table, spreadsheets
text
graph
pdf
email

BENCH:
shell
python
pandas
spreadsheet
modules

DEEP:
linear systems
tuple theory
matrix algebra

OTHER:
exception handling
type checking
file encoding

SPACES:
raw binary space - unix terminal util
forge of logic engines - c

logic of situation - perl python
spreadsheet information logic - pandas
matrix numerical information - numpy
multiset ledger reconcile information
graphic projection - data plotting
report processing - pdf
stream of data - data is not retained, only shapes a image state or alerts upon event
structural/related tabular data, unisolated structures - sql, trees

Programming Theory:
Blackbox Abstraction
Longhand

Programming falls into two schools of reasoning.
Bluecollar usage of general purpose tools.
Whitecollar construction of tools.

Accumulation of expertise of the one does not accummulate the other.
These are two independent paths which establish contrary frames of existence.
Proficient manipulation VS hacking creation.

Blackbox Abstractions allow war tested immediate modules to process the GENERAL situation. The tool of use lives in an entire ecosystem of the module.
This nested entity is not easily modified, less so understood.

Longhand keeps the hands of the creator in the clay. the logic is bare, there exists a flow and readership. Portability is supreme because the logic is explicitly simple.
The code is long. what may be a simple call in a Blackbox, may be the most complex machinery in longhand.

Blackbox Abstraction builds the bluecollar worker mindset: exist within the acceptable realm, and be entirely powerless when the situation does not fit.

Longhand builds the whitecollar mindset: exist entirely unlimited except by time. Powerful in all situations.